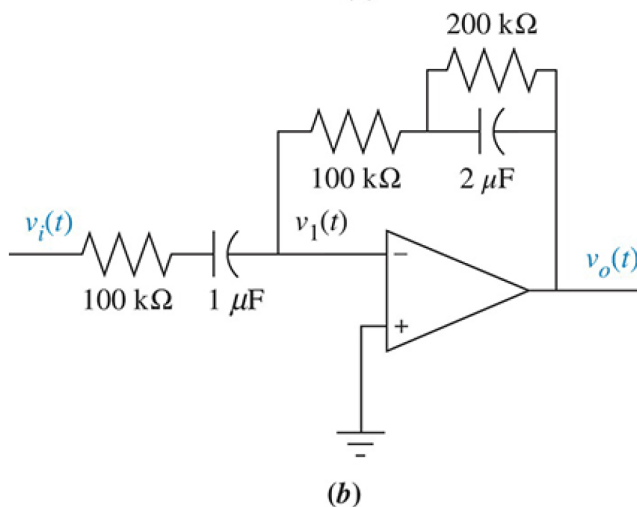
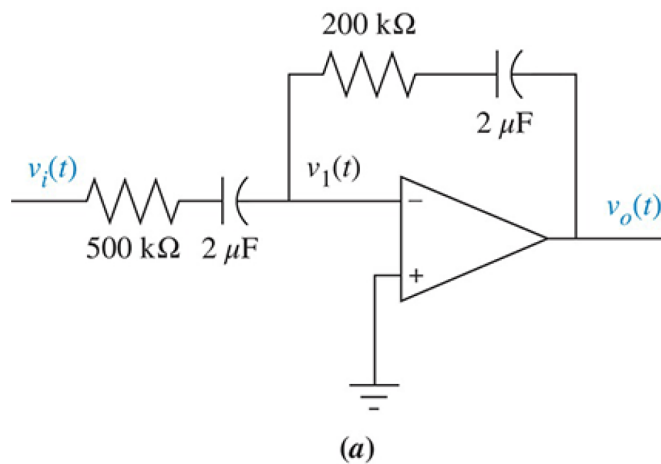


Midterm Review

2026年2月8日 9:52

Midterm复习:

17. For each of the circuits shown in [Figure P2.5](#), find the corresponding transfer function $G(s) = V_o(s)/V_i(s)$. [[Section: 2.4](#)]



Midterm Review

Q17.

$$G = \frac{V_o(s)}{V_i(s)}$$

2.

$$\therefore G = -\frac{Z_f}{Z_i} = \text{gain}$$

$$Z_f = 200 \times 10^3 + \frac{1}{2s} = 2 \times 10^5 + \frac{1}{2 \times 10^{-6} \times s}$$

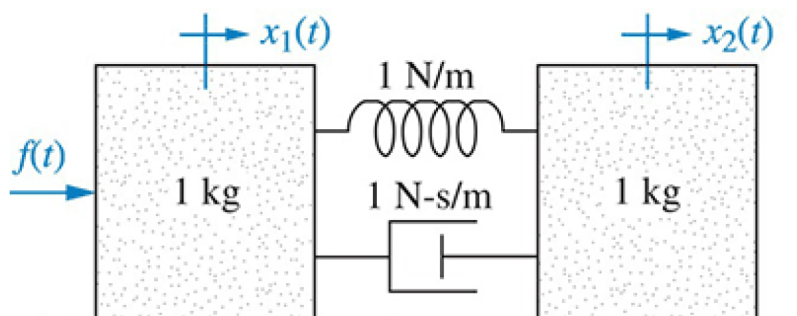
$$= \frac{4 \times 10^{-1} \times s + 1}{2 \times 10^{-6} \times s}$$

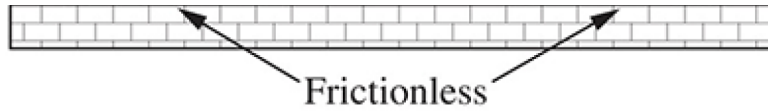
$$Z_i = 500 \times 10^3 + \frac{1}{2s} = 5 \times 10^5 + \frac{1}{2 \times 10^{-6} \times s} = \frac{10 \times 10^{-1} \times s + 1}{2 \times 10^{-6} \times s}$$

$$= \frac{s + 1}{2 \times 10^{-6} \times s}$$

$$\therefore G = -\frac{Z_f}{Z_i} = -\frac{\frac{0.4s + 1}{2 \times 10^{-6} \times s}}{\frac{s + 1}{2 \times 10^{-6} \times s}} = -\frac{0.4s + 1}{s + 1}$$

SS 20. Find the transfer function, $G(s) = X_2(s)/F(s)$, for the translational mechanical network shown in [Figure P2.8](#). [[Section: 2.5](#)]

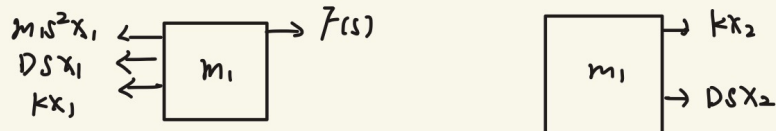




Q20.

$$G(s) = \frac{x_2(s)}{F(s)}, \quad m_1 = m_2 = 1 \text{ kg}, \quad k=1, \quad D=1$$

$\rightarrow x_1$

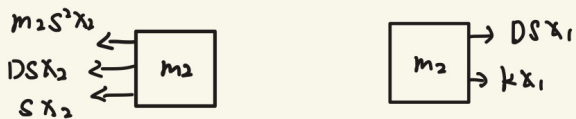


$$\therefore m_1 s^2 x_1 + D s x_1 + k x_1 = F(s) + k x_2 + D s x_2$$

$$\therefore s^2 x_1 + s x_1 + x_1 = F(s) + x_2 + s x_2$$

$$(s^2 + s + 1) x_1 - (s + 1) x_2 = F(s)$$

$\rightarrow x_2$



$$\therefore m_2 s^2 x_2 + D s x_2 + k x_2 = D s x_1 + k x_1$$

$$\therefore s^2 x_2 + s x_2 + x_2 - (s + 1) x_1 = 0$$

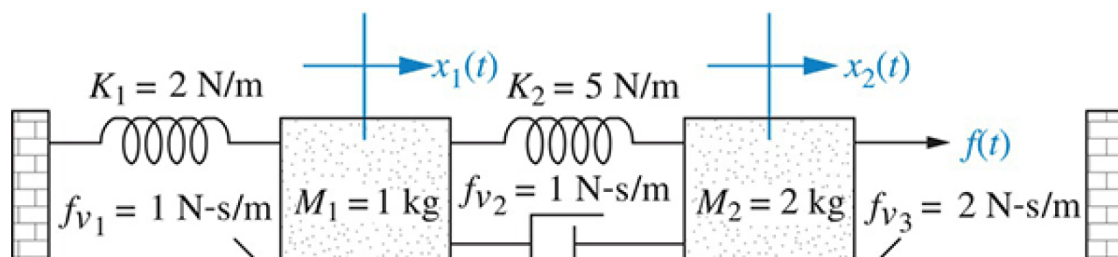
$$\therefore \begin{bmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & s^2 + s + 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$\therefore x_2 = \frac{\begin{vmatrix} s^2 + s + 1 & F(s) \\ -(s + 1) & 0 \end{vmatrix}}{\begin{vmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & s^2 + s + 1 \end{vmatrix}}$$

$$\begin{vmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & s^2 + s + 1 \end{vmatrix}$$

$$x_2 = \frac{F(s+1)}{s^4 + 2s^3 + 2s^2} \quad \therefore G = \frac{s+1}{s^4 + 2s^3 + 2s^2}$$

21. Find the transfer function, $G(s) = X_2(s)/F(s)$, for the system shown in [Figure P2.9](#) [[Section: 2.5](#)]



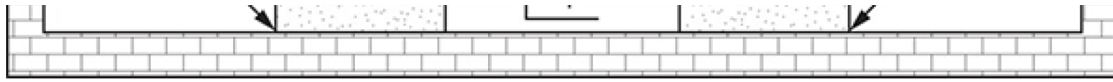


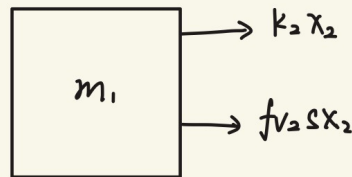
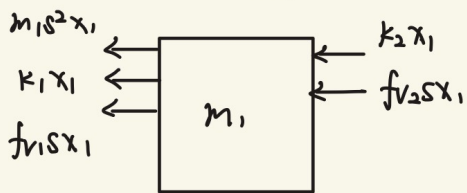
FIGURE P2.9

Q21

$$G(s) = \frac{X_2(s)}{F(s)}$$

$$k_1=2, f_{V1}=1, m_1=1, k_2=5, f_{V2}=1, m_2=2, f_{V3}=2$$

$\rightarrow x_1$

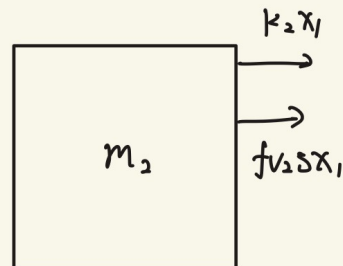
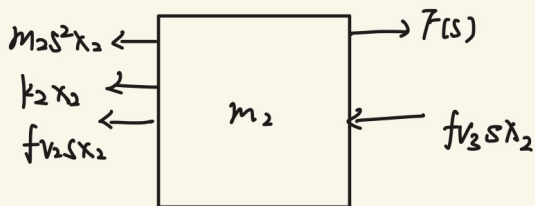


$$\therefore m_1 s^2 x_1 + k_1 x_1 + f_{V1} s x_1 + k_2 x_1 + f_{V2} s x_1 = k_2 x_2 + f_{V2} s x_2$$

$$(s^2 + 2 + s + 5 + s) x_1 - (5 + s) x_2 = 0$$

$$(s^2 + 2s + 7) x_1 - (s + 5) x_2 = 0$$

$\rightarrow x_2$



$$m_2 s^2 x_2 + k_2 x_2 + f_{V2} s x_2 + f_{V3} s x_2 = F(s) + k_2 x_1 + f_{V2} s x_1$$

$$\Rightarrow 2s^2 x_2 + 5x_2 + s x_2 + 2s x_2 = F(s) + 5x_1 + s x_1$$

$$(2s^2 + 3s + 5) x_2 - (s + 5) x_1 = F(s)$$

$$-(s + 5) x_1 + (2s^2 + 3s + 5) x_2 = F(s)$$

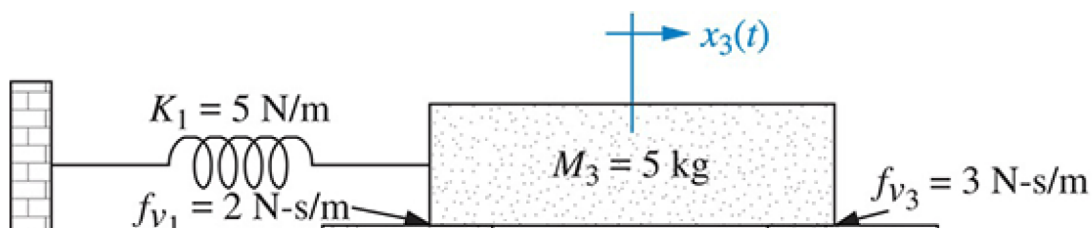
$$\begin{bmatrix} s^2 + 2s + 7 & -(s + 5) \end{bmatrix} \begin{bmatrix} x_1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

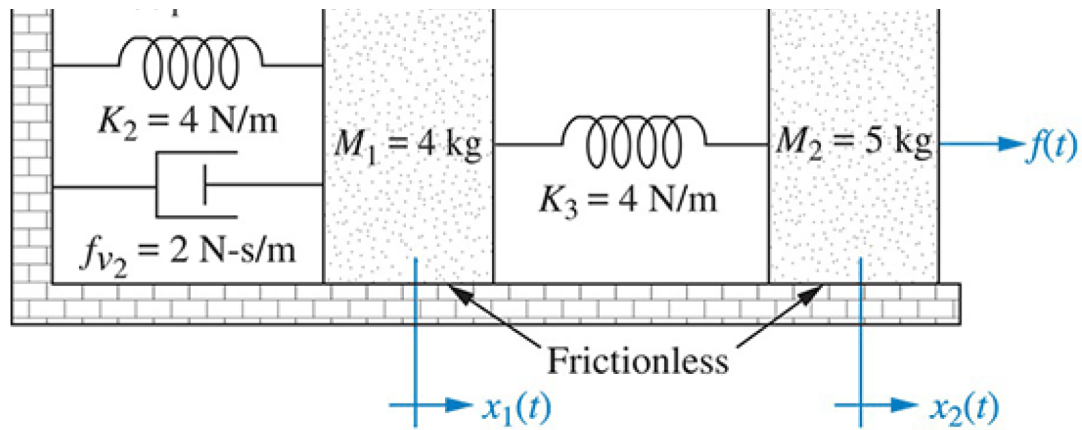
$$\begin{bmatrix} -(s+5) & 2s^2+3s+5 \\ s^2+2s+7 & 0 \\ -(s+5) & F(s) \end{bmatrix} \begin{bmatrix} x_2 \end{bmatrix} = \begin{bmatrix} F(s) \end{bmatrix}$$

$$X_2 = \frac{\begin{vmatrix} s^2+2s+7 & 0 \\ -(s+5) & F(s) \end{vmatrix}}{\begin{vmatrix} s^2+2s+7 & -(s+5) \\ -(s+5) & 2s^2+3s+5 \end{vmatrix}} = \frac{F(s) \cdot (s^2+2s+7)}{2s^4+7s^3+24s^2+21s+10}$$

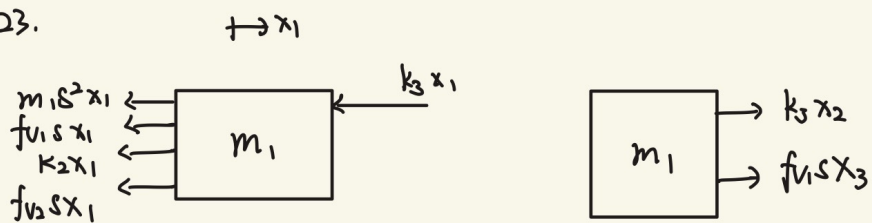
$$\therefore G(s) = \frac{(s^2+2s+7)}{2s^4+7s^3+24s^2+21s+10}$$

23. Write, but do not solve, the equations of motion for the translational mechanical system shown in [Figure P2.11](#). [[Section: 2.5](#)]



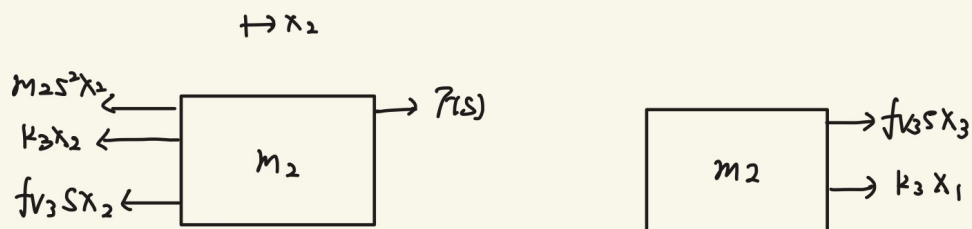


Q23.



$$\therefore m_1 s^2 x_1 + f_{v1} s x_1 + k_2 x_1 + f_{v2} s x_1 + k_3 x_1 = k_3 x_2 + f_{v1} s x_3$$

$$\therefore (4s^2 + 2s + 4 + 2s + 4)x_1 - 4x_2 - 2s x_3 = 0$$



$$\therefore m_2 s^2 x_2 + k_3 x_2 + f_{v3} s x_2 = F(s) + f_{v3} s x_3 + k_3 x_1$$

$$5s^2 x_2 + 4x_2 + 3s x_2 = F(s) + 3s x_3 + 4x_1$$

$$-4x_1 + (5s^2 + 4 + 3s)x_2 - 3s x_3 = F(s)$$





$$f_{V3} s x_3$$

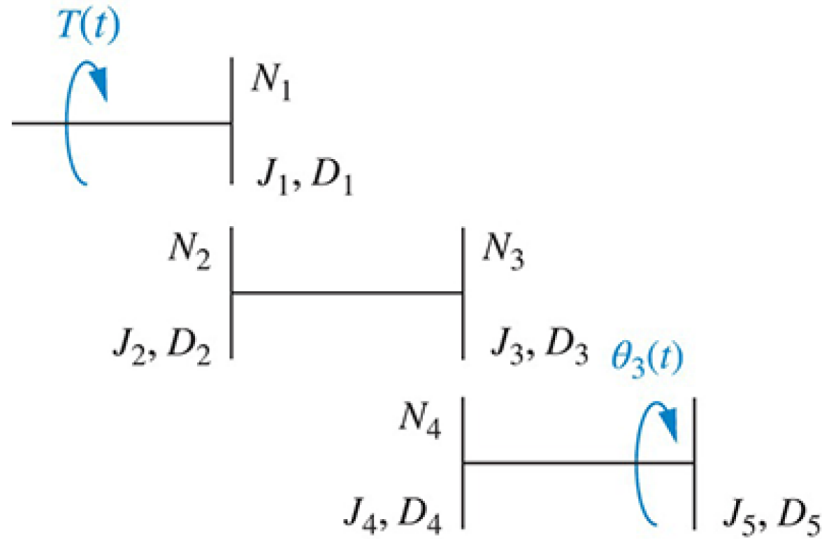
$$m_3 s^2 x_3 + k_1 x_3 + f_{V1} s x_3 + f_{V3} s x_3 = f_{V1} s x_1 + f_{V3} s x_2$$

$$5s^2 x_3 + 5x_3 + 2s x_3 + 3s x_3 = 2s x_1 + 3s x_2$$

$$\therefore 2s x_1 + (5s^2 + 5s + 5)x_3 - 3s x_2 = 0$$

27. For the rotational mechanical system with gears shown in Figure P2.15, find the transfer function, $G(s) = \theta_1(s)/T(s)$. The

gears have inertia and bearing friction as shown. [Section: 2.7]



Q27.

$$G(s) = \frac{\theta_3(s)}{T(s)}$$

$$J_e = J_1 \cdot \frac{N_2^2}{N_1^2} \cdot \frac{N_4^2}{N_3^2} + J_2 \cdot \frac{N_4^2}{N_3^2} + J_3 \cdot \frac{N_4^2}{N_3^2} + J_4 + J_5$$

$$D_e = D_1 \cdot \frac{N_2^2}{N_1^2} \cdot \frac{N_4^2}{N_3^2} + (D_2 + D_3) \frac{N_4^2}{N_3^2} + D_4 + D_5$$

$$T_e : \because \frac{T_2}{T_1} = \frac{N_2}{N_1} \therefore T_2 = \frac{N_2}{N_1} T_1, \quad T_3 = T_e = \frac{N_4}{N_3} \frac{N_2}{N_1} T(s)$$

$$\therefore J_e s^2 \theta_3 + D_e s \theta_3 = \frac{N_4}{N_3} \frac{N_2}{N_1} T(s)$$

$$G = \frac{\theta_3}{T}$$

29. Obtain the transfer function, $G(s) = \theta_2(s)/T(s)$, for the system of Figure P2.17. [Section: 2.7]

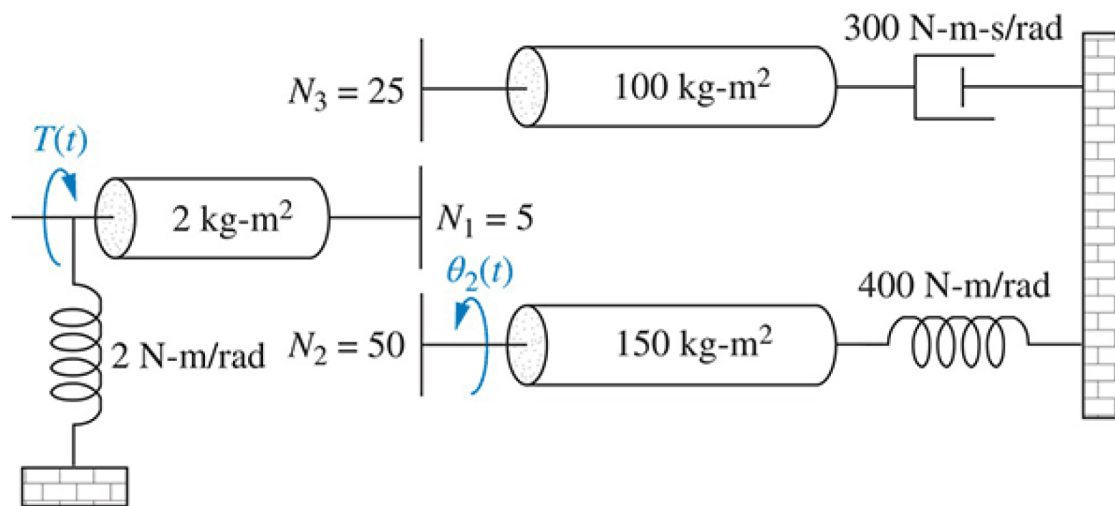


FIGURE P2.17

29.

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

$$J_e = 150 + 2 \times \frac{N_2^2}{N_1^2} + 100 \times \frac{N_1^2}{N_2^2} \times \frac{N_2^2}{N_1^2}$$

$$= 150 + 2 \times \frac{50^2}{5^2} + 100 \times \frac{50^2}{25^2}$$

$$= 150 + 200 + 400$$

$$= 750$$

$$D_e = 300 \times \frac{N_2^2}{N_3^2} = 300 \times \frac{50^2}{25^2} = 1200$$

$$k_e = 2 \times \frac{N_2^2}{N_1^2} + 400 = 2 \times \frac{50^2}{5^2} + 400 = 600$$

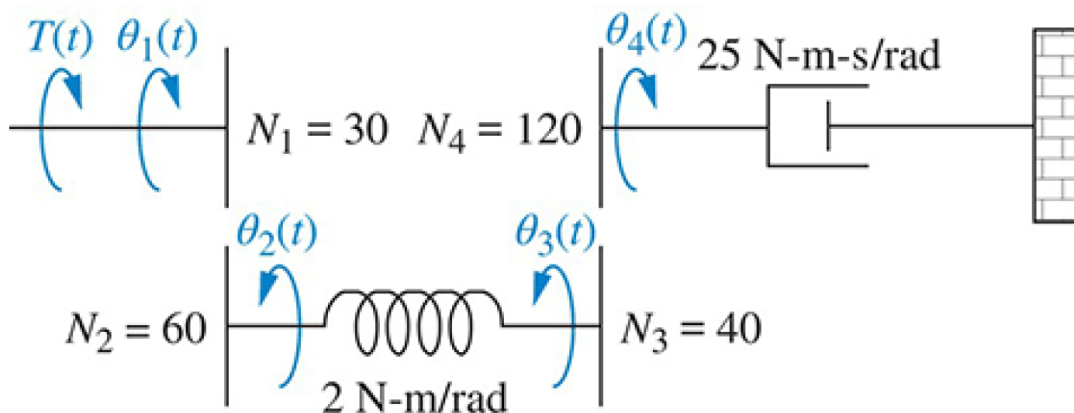
$$\therefore \frac{T_2}{T_1} = \frac{N_2}{N_1} \quad \therefore T_2 = \frac{N_2}{N_1} T_1(s) = 10 T_1(s)$$

$$\therefore 750s^2 \theta_2 + 1200s \theta_2 + 600 \theta_2 = 10 T_1(s)$$

$$\therefore \theta_2 = \frac{10 T_1(s)}{750s^2 + 1200s + 600} = \frac{T_1(s)}{75s^2 + 120s + 60}$$

$$\therefore G(s) = \frac{1}{75s^2 + 120s + 60}$$

30. For the rotational system of [Figure P2.18](#), find the transfer function, $G(s) = \theta_2(s)/T(s)$. [[Section: 2.7](#)]



Ass.

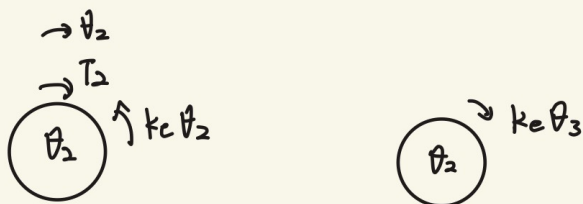
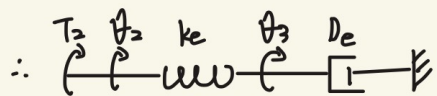
$$G(s) = \frac{\theta_2(s)}{T(s)}$$

$$J_e = 0$$

$$D_e = 25 \times \frac{N_3^2}{N_4^2} = 2.78$$

$$K_e = 2$$

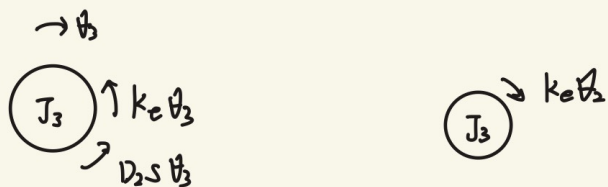
$$\frac{T_1}{T_2} = \frac{N_1}{N_2} \therefore T_2 = \frac{N_2}{N_1} T_1 = 2T(s)$$



$$\therefore K_e \theta_2 = T_2 + K_e \theta_3$$

$$2\theta_2 = 2T(s) + 2\theta_3$$

$$\therefore 2\theta_2 - 2\theta_3 = 2T(s)$$



$$K_e \theta_3 + D_e s \theta_3 = K_e \theta_2$$

$$2\theta_3 + 2.78s\theta_3 = 2\theta_2$$

$$\therefore -2\theta_2 + (2.78s + 2)\theta_3 = 0$$

$$\therefore \begin{bmatrix} 2 & -2 \\ 0 & 2.78s + 2 \end{bmatrix} \begin{bmatrix} \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 2T \\ 0 \end{bmatrix}$$

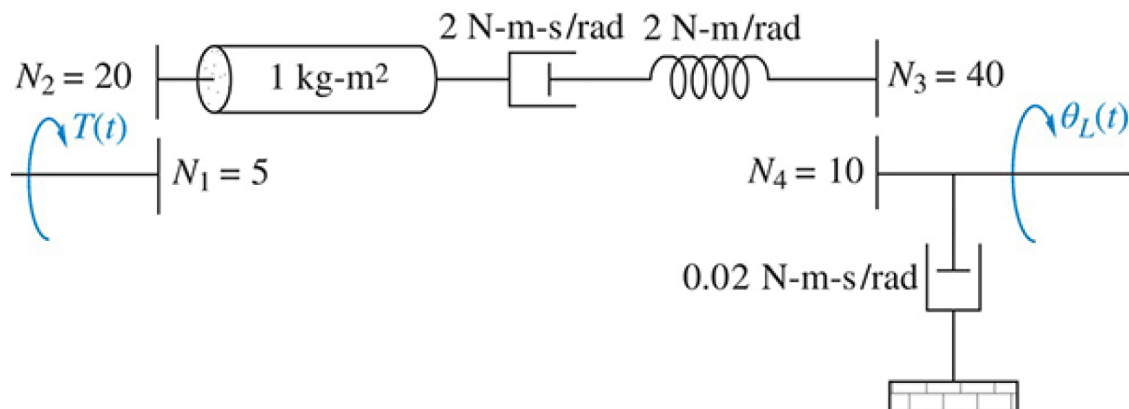
$$\begin{bmatrix} -2 & 2.78s+2 \end{bmatrix} \begin{bmatrix} \theta_3 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\therefore \theta_2(s) = \frac{\begin{vmatrix} 27 & -2 \\ 0 & 2.78s+2 \end{vmatrix}}{\begin{vmatrix} 2 & -2 \\ -2 & 2.78s+2 \end{vmatrix}} = \frac{5.56s+47}{5.56s}$$

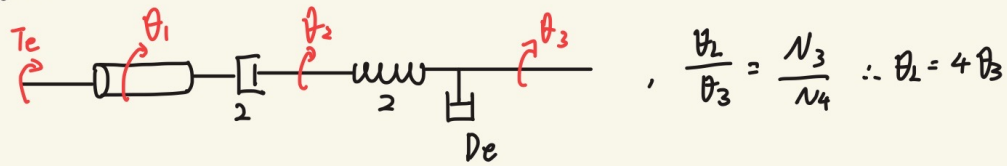
$$= 7 + 0.72 \frac{7}{s}$$

$$\therefore G(s) = 1 + \frac{0.72}{s} = \frac{s+0.72}{s}$$

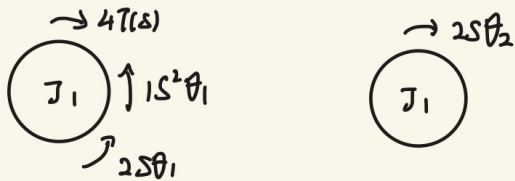
31. For the rotational system shown in [Figure P2.19](#), find the transfer function, $G(s) = \theta_L(s)/T(s)$. [[Section: 2.7](#)]



Q31.

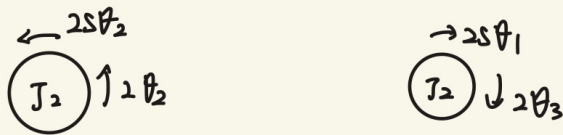


$$T_c = \frac{N_2}{N_1} T = 4 T(s) \quad , \quad D_c = \frac{N_3^2}{N_4^2} \times 0.02 = 16 \times 0.02 = 0.32$$



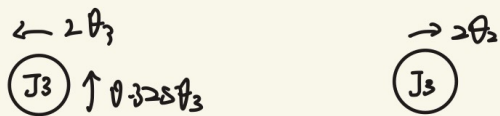
$$\therefore 1s^2\theta_1 + 2s\theta_1 = 4T(s) + 2s\theta_2$$

$$\therefore (s^2 + 2s)\theta_1 - 2s\theta_2 = 4T(s)$$



$$\therefore 2s\theta_2 + 2\theta_3 = 2s\theta_1 + 2\theta_3$$

$$\therefore 2s\theta_1 - (2s+2)\theta_2 + 2\theta_3 = 0$$



$$2\theta_3 + 0.32s\theta_3 = 2\theta_2$$

$$\therefore -2\theta_2 + (0.32s+2)\theta_3 = 0$$

$$\begin{bmatrix} s^2+2s & -2s & 0 \\ 2s & -(2s+2) & 2 \\ 0 & -2 & 0.32s+2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 4T(s) \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & 0.32s+2 \end{bmatrix} \begin{bmatrix} \theta_3 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\theta_3 = \frac{100T}{4s^3 + 29s^2 + 8s}$$

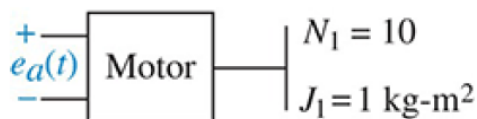
$$\therefore \frac{\theta_2}{\theta_3} = \frac{N_3}{N_4} \quad \therefore \theta_2 = \frac{N_3}{N_4} \theta_3 = \frac{400T}{4s^3 + 29s^2 + 8s}$$

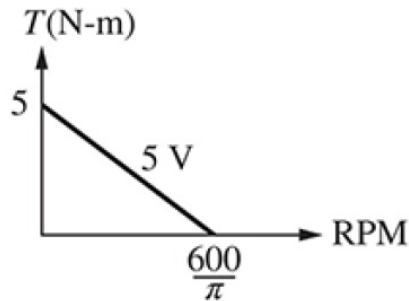
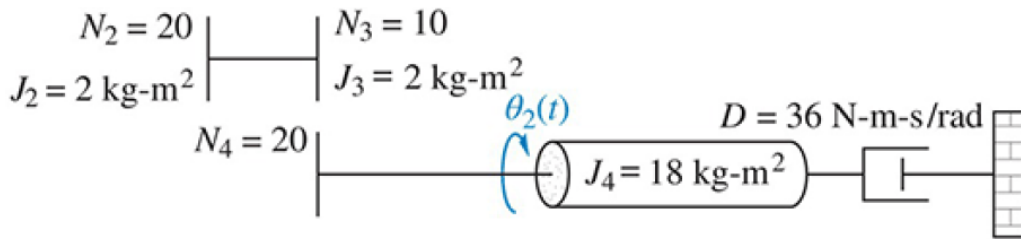
$$\therefore G(s) = \frac{\theta_2}{T} = \frac{400}{4s^3 + 29s^2 + 8s}$$

SS 35. The motor whose torque–speed characteristics are shown in [Figure P2.23](#) drives the load shown in the diagram. Some of the

gears have inertia. Find the transfer function, $G(s) = \theta_2(s)/E_a(s)$.

[[Section: 2.8](#)]





Q35.

$$G(s) = \frac{\theta_2(s)}{E_a(s)}, \quad \text{Rad/s} = \frac{2\pi}{60} \text{ rpm}$$

$$\therefore T_m = -\frac{k_b k_v}{R_a} \omega_m + \frac{k_v}{R_a} e_a$$

From the graph: $e_a = 5 \text{ V}$, when $\omega = 0$, $T = 5$

$$\therefore 5 = \frac{k_v}{R_a} \times 5 \quad \therefore \frac{k_v}{R_a} = 1$$

$$\text{When } \omega_m = \frac{600}{\pi} \times \frac{2\pi}{60} = 20 \text{ rad/s}, \quad T = 0$$

$$\therefore -\frac{k_b k_t}{R_a} \times 20 + 5 = 0$$

$$\therefore \frac{k_b k_t}{R_a} = \frac{5}{20} = \frac{1}{4}$$

$$\therefore \frac{\theta_m}{E_a} = \frac{\frac{k_t}{R_a} \cdot \frac{1}{J_e}}{s \left[s + \frac{1}{J_e} \left(D + \frac{k_b k_t}{R_a} \right) \right]}, \quad \theta_2 = \frac{\theta_m}{4} \quad \therefore \theta_m = 4\theta_2$$

$$J_e = J_4 + \frac{N_4^2}{N_3^2} (J_3 + J_2) + J_1 \frac{N_4^2}{N_2^2} \frac{N_2^2}{N_1^2}$$

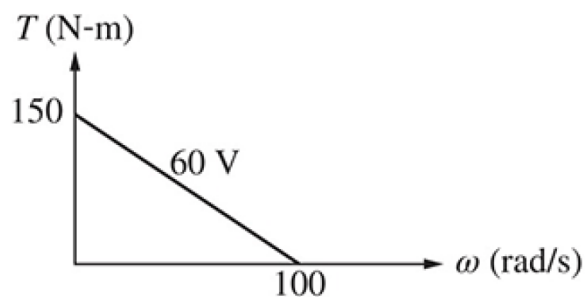
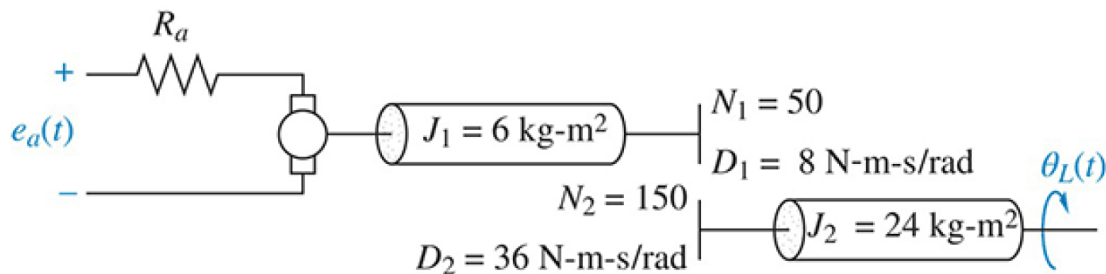
$$= 18 + 4(2+2) + 1 \times 4 \times 4$$

$$= 18 + 16 + 16 = 50$$

$$D_e = 36$$

$$\therefore G(s) = \frac{\frac{4}{50}}{s[s + \frac{1}{50}(36 + \frac{1}{4})]} = \frac{4}{50s^2 + 31.25s}$$

34. In [Figure P2.22](#), a load is driven with a motor whose torque–speed characteristic is shown in the Figure. Determine the transfer function, $G(s) = \theta_L(s)/E_a(s)$



Q34.

$$G(s) = \frac{\theta_L(s)}{E_a(s)}, \quad V_b = k_b s \theta_m, \quad T_m = k_t I_a, \quad I_a = \frac{T_m}{k_t}$$

$$\therefore E_a(s) - I_a R_a - k_b s \theta_m = 0$$

$$E_a - \frac{T_m}{k_t} R_a - k_b s \theta_m = 0 \Rightarrow T_m = \frac{k_t}{R_a} E_a - \frac{k_b k_t}{R_a} s \theta_m$$

$$\therefore T_m = -\frac{k_b k_t}{R_a} s \theta_m + \frac{k_t}{R_a} e_a$$

When $W=100$, $T=0$; $W=0$, $T=150$

$$\therefore 150 = \frac{k_b}{R_a} \times 60 \quad \therefore \frac{k_b}{R_a} = \frac{15}{6}$$

$$0 = -\frac{k_b k_c}{R_a} \times 100 + \frac{15}{6} \times 60$$

$$\frac{k_b k_c}{R_a} = \frac{150}{100} = 1.5 = \frac{3}{2}, \quad k_b = \frac{3}{2} \times \frac{6}{15} = \frac{3}{5}$$

$$J_c = J_2 + J_1 \cdot \frac{N_2^2}{N_1^2} = 24 + 6 \times \frac{150^2}{50^2} = 78$$

$$D_c = 36 + 8 \times \frac{150^2}{50^2} = 108$$

$$T_c = \frac{150}{50} \times T(5) = 3T(5)$$

$$\therefore 78s^2 \theta_L + 108s \theta_L = 3T \quad (2) \quad \theta_L = \frac{3T}{78s^2 + 108s}$$

$$\therefore \frac{\theta_m}{\theta_L} = \frac{N_2}{N_1} \quad \therefore \theta_m = 3\theta_L$$

$$\therefore T_m = \frac{15}{6} E_a - \frac{3}{2}s \theta_m = \frac{15}{6} E_a - \frac{9}{2}s \theta_L$$

$$\therefore \theta_L = \frac{3 \left(\frac{15}{6} E_a - \frac{9}{2}s \theta_L \right)}{78s^2 + 108s}$$

$$\theta_L = \frac{5E_a}{52s^2 + 81s} \quad \therefore G(s) = \frac{\theta_L}{E_a} = \frac{5}{52s^2 + 81s}$$

50. In a paint mixing plant, two tanks supply fluids to a mixing cistern. The height, h , of the fluid in the cistern is dependent upon the difference between the input mass flow rate, q , and the output flow rate, q_e . A nonlinear differential equation describing this dependency is given by (Schiop, 2010)

$$\frac{dh}{dt} + \frac{A_e}{A} \sqrt{2gh} = \frac{q}{\rho A}$$

where A = cross-sectional area of the cistern, A_e = cross-sectional area of the exit pipe, g = acceleration due to gravity, and ρ = liquid density.

a. Linearize the nonlinear equation about the equilibrium point (h_o, q_o) and find the transfer function relating the output cistern fluid level, $H(s)$, to the input mass flow rate, $Q(s)$.

b. The color of the liquid in the cistern can be kept constant

D. The color of the liquid in the cistern can be kept constant by adjusting the input flow rate, q , assuming the input flow's color is specifically controlled. Assuming an average height,

h_{av} , of the liquid in the cistern, the following equation relates the net flow of color to the cistern to the color in the cistern.

$$e_1 q - e q_e = \frac{d}{dt}(\rho A e h_{av})$$

where e_1 = fractional part of flow representing color into the cistern, and e = fractional part of the cistern representing color in the cistern. Assume that the flow out of the cistern is constant and use the relationship, $q_e = \rho A_e \sqrt{2gh_{av}}$, along with the given equation above to find the transfer function, $E(s)/Q(s)$, that relates the color in the cistern to the input flow rate.

Q50.

$$\text{Let } h = h_0 + \Delta h, \quad q = q_0 + \Delta q$$

$$\frac{d(h_0 + \Delta h)}{dt} + \frac{A_e}{A} \sqrt{2gh} = \frac{q_0 + \Delta q}{\rho A}$$

$$\downarrow$$

$$\frac{d(\Delta h)}{dt}$$

$$\sqrt{2gh} = \sqrt{2g(h_0 + \Delta h)}, \quad \sqrt{2g(h_0 + \Delta h)} - \sqrt{2g(h_0)} = \left. \frac{d\sqrt{2gh}}{dh} \right|_{h=h_0} \cdot \Delta h$$

$$\therefore \sqrt{2gh} = \sqrt{2gh_0} + \Delta h \cdot \frac{1}{2} (2gh_0)^{-\frac{1}{2}} \cdot 2g$$

$$= \sqrt{2gh_0} + \frac{g}{\sqrt{2gh_0}} \Delta h$$

$$\therefore \text{DE is } \frac{d(\Delta h)}{dt} + \frac{A_e}{A} \frac{g}{\sqrt{2gh_0}} \Delta h = \frac{\Delta q}{\rho A}$$

$$sH(s) + \frac{Ae g}{A\sqrt{2gh_0}} H(s) = \frac{Q(s)}{PA}$$

$$H\left(s + \frac{Ae g}{A\sqrt{2gh_0}}\right) = \frac{Q}{PA}$$

$$\therefore G = \frac{H}{Q} \quad \therefore G(s) = \frac{1}{\left(s + \frac{Ae g}{A\sqrt{2gh_0}}\right) PA}$$

$$(b) G(s) = \frac{E(s)}{Q(s)}$$

$$e_1 q - e PAe\sqrt{2gh_{av}} = \frac{d}{dt}(PAeh_{av})$$

$$= PAh_{av} \cdot \frac{de}{dt}$$

$$PAh_{av} \frac{de}{dt} + PAe\sqrt{2gh_{av}} e = e_1 q$$

$$PAh_{av} s E(s) + PAe\sqrt{2gh_{av}} E(s) = e_1 Q(s)$$

$$G(s) = \frac{e_1}{PAh_{av}s + PAe\sqrt{2gh_{av}}}$$

SS 41. Consider the differential equation

$$\frac{d^3 x}{dt^3} + 10 \frac{d^2 x}{dt^2} + 20 \frac{dx}{dt} + 15x = f(x)$$

where $f(x)$ is the input and is a function of the output, x . If $f(x) = 3e^{-5x}$, linearize the differential equation for x near 0.
[[Section: 2.10](#)]

Q 41.

$$\frac{d^3 x}{dt^3} + 10 \frac{d^2 x}{dt^2} + 20 \frac{dx}{dt} + 15x = 3e^{-5x}$$

$$\text{Let } x = x_0 + \delta x = 0 + \delta x$$

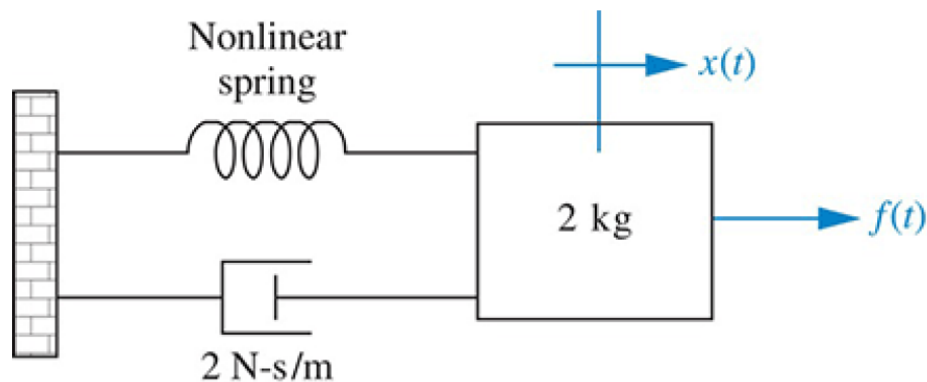
$$\text{and: } f(x) - f(x_0) = \delta x \cdot f'(x_0) \quad , \quad f'(x) = -15e^{-5x}$$

$$\begin{aligned} \therefore 3e^{-5x} - 3e^{-5x_0} &= \delta x \cdot (-15e^{-5 \cdot 0}) \\ &= -15 \delta x \end{aligned}$$

$$\therefore 3e^{-5x} = 3 - 15 \delta x$$

$$\therefore \frac{d^3(\delta x)}{dt^3} + 10 \frac{d^2(\delta x)}{dt^2} + 20 \frac{d(\delta x)}{dt} + 15 \delta x = 3 - 15 \delta x$$

42. For the translational mechanical system with a nonlinear spring shown in [Figure P2.25](#), find the transfer function, $G(s) = X(s)/F(s)$, for small excursions around $f(t) = 1$. The spring is defined by $x_s(t) = 1 - e^{-f_s(t)}$, where $x_s(t)$ is the spring displacement and $f_s(t)$ is the spring force. [[Section: 2.10](#)]



Q42.

$$G(s) = \frac{X(s)}{F(s)}, \quad f(t) = 1, \quad x_s(t) = 1 - e^{-f_s(t)}$$

$$\therefore F = kx, \quad e^{-f_s(t)} = 1 - x_s(t), \quad -f_s(t) = \ln(1 - x_s(t))$$

$$f_s(t) = -\ln(1 - x_s(t))$$

$$f_s'(t) = \frac{1}{1 - x_s(t)}$$

$$\therefore 2 \frac{d^2 x}{dt^2} + 2 \frac{dx}{dt} + f_s(t) = f(t)$$

① Find equilibrium point :

$$\text{Let } \dot{x}(t) = 0 \text{ and } f(t) = 1 \quad \therefore -\ln(1 - x) = 1, \quad 1 - x = e^{-1} \\ x_0 = 1 - e^{-1}$$

② Linearization :

$$\text{Let } x = x_0 + \delta x, \quad \therefore f_s(t) = -\ln(1 - x_0 - \delta x)$$

$$\therefore f_s(x) - f_s(x_0) = \delta x \cdot f_s'(x_0) \quad \therefore -\ln(1 - x) + \ln(1 - x_0) = \delta x \cdot \frac{1}{1 - x_0}$$

$$\therefore f_s(x) = -\ln(1 - 1 + e^{-1}) + \delta x \cdot \frac{1}{1 - 1 + e^{-1}}$$

$$= 1 + \delta x \cdot e = e \delta x$$

$$\therefore DE = 2 \frac{d^2(\delta x)}{dt^2} + 2 \frac{d(\delta x)}{dt} + e \delta x = f(t)$$

$$2s^2 X(s) + 2s X(s) + e X(s) = F(s)$$

$$\therefore G(s) = \frac{X(s)}{F(s)} \quad \therefore \frac{X(s)}{F(s)} = \frac{1}{2s^2 + 2s + e} = G(s)$$

1. Electric Circuits (25 marks) (By hand & By MATLAB)

Note: A warning in advance - the algebraic substitutions can get really messy. You should use MATLAB to help with simplifying the expression and solving for determinants. You must show all matrices and vectors by hand to receive full marks for this question.

- (a) (15 marks) Use mesh analysis to find the transfer function $H(s) = \frac{V_o(s)}{V_s(s)}$ of the circuit shown below:

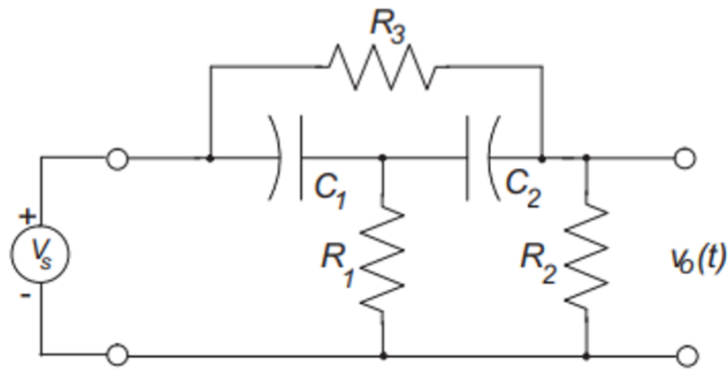


Figure 1a

- (b) (10 marks) A “wheatstone bridge” is used in many sensors in control and measurement engineering. For example the elements in the bridge shown below might be sensitive to strain or chemicals in the environment.

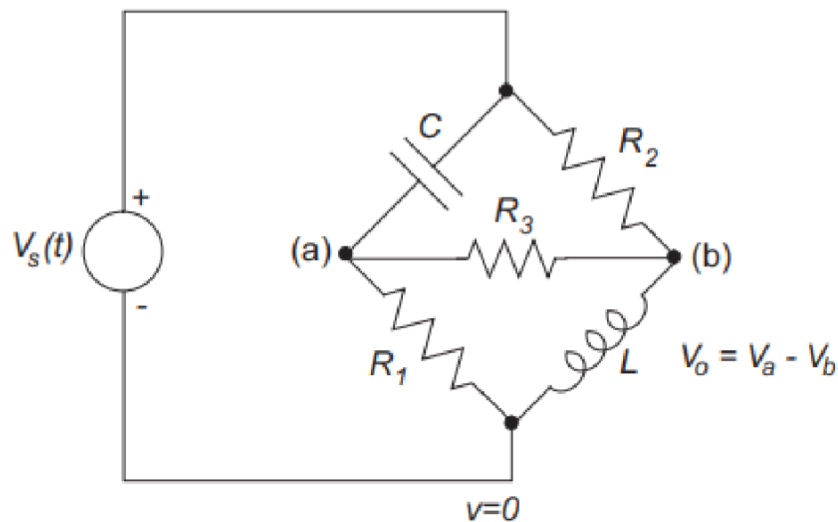


Figure 1b

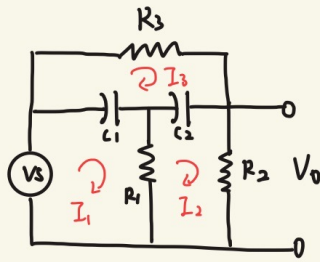
A voltage source is connected across the bridge, and the output V_o is taken as the voltage difference between the nodes (a) and (b) as shown, that is

$$V_o = V_a - V_b$$

Your task is to derive the transfer function relating $V_o(s)$ to the source voltage $V_s(s)$ using nodal analysis.

Q1. (a)

$$G(s) = \frac{V_o(s)}{V_s(s)} \quad , \quad V_o = I_2 R_2$$



$$Z_{C1} = \frac{1}{C_1 s}, \quad Z_{C2} = \frac{1}{C_2 s}$$

$$Z_{1,2,3} = R_{1,2,3}$$

$$1. \quad V_s - \frac{1}{C_1 s} (I_1 - I_3) - R_1 (I_1 - I_2) = 0$$

$$\left(\frac{1}{C_1 s} + R_1\right) I_1 - R_1 I_2 - \frac{1}{C_1 s} I_3 = V_s \quad (1)$$

2.

$$0 - \frac{1}{C_2 s} (I_2 - I_3) - I_2 R_2 - R_1 (I_2 - I_1) = 0$$

$$\frac{1}{C_2 s} (I_2 - I_3) + I_2 R_2 + R_1 (I_2 - I_1) = 0$$

$$-R_1 I_1 + \left(\frac{1}{C_2 s} + R_2 + R_1\right) I_2 - \frac{1}{C_2 s} I_3 = 0 \quad (2)$$

3.

$$0 - I_3 R_3 - \frac{1}{C_2 s} (I_3 - I_2) - \frac{1}{C_1 s} (I_3 - I_1) = 0$$

$$I_3 R_3 + \frac{1}{C_2 s} (I_3 - I_2) + \frac{1}{C_1 s} (I_3 - I_1) = 0$$

$$-\frac{1}{C_1 s} I_1 - \frac{1}{C_2 s} I_2 + \left(R_3 + \frac{1}{C_2 s} + \frac{1}{C_1 s}\right) I_3 = 0 \quad (3)$$

$$\begin{bmatrix} \frac{1}{C_1 s} + R_1 & -R_1 & -\frac{1}{C_1 s} \\ -R_1 & \frac{1}{C_2 s} + R_1 + R_2 & -\frac{1}{C_2 s} \\ -\frac{1}{C_1 s} & -\frac{1}{C_2 s} & R_3 + \frac{1}{C_2 s} + \frac{1}{C_1 s} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \end{bmatrix}$$

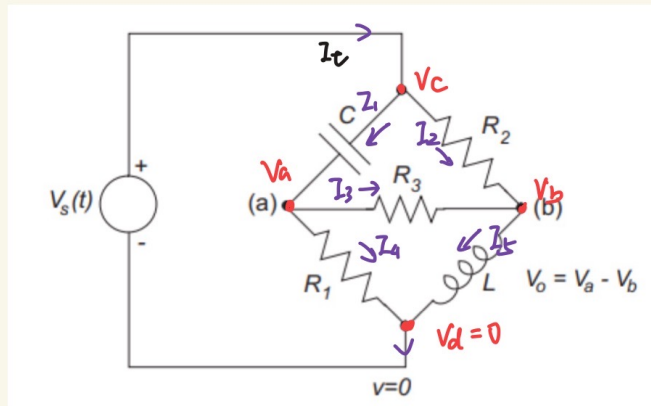
$$\begin{vmatrix} \frac{1}{C_1 s} + R_1 & V_s & -\frac{1}{C_1 s} \\ -R_1 & 0 & -\frac{1}{C_2 s} \\ -\frac{1}{C_1 s} & 0 & R_3 + \frac{1}{C_2 s} + \frac{1}{C_1 s} \end{vmatrix}$$

$$I_2 = \frac{1}{\begin{vmatrix} \frac{1}{C_1 s} + R_1 & -R_1 & -\frac{1}{C_1 s} \\ -R_1 & \frac{1}{C_2 s} + R_1 + R_2 & -\frac{1}{C_2 s} \\ \frac{-1}{C_1 s} & \frac{-1}{C_2 s} & R_3 + \frac{1}{C_2 s} + \frac{1}{C_1 s} \end{vmatrix}}$$

$$\therefore G(s) = \frac{V_o}{V_s}, \quad V_o = I_2 R_2 \quad \therefore G(s) = \frac{I_2 R_2}{V_s}$$

$$G(s) = \frac{R_1 R_2 R_3 C_1 C_2 s^2 + (R_1 R_2 C_1 + R_1 R_2 C_2) s + R_2}{R_1 R_2 R_3 C_1 C_2 s^2 + (R_1 R_2 C_1 + R_1 R_2 C_2 + R_1 R_3 C_1 + R_1 R_3 C_2 + R_2 R_3 C_2) s + R_2 + R_3}$$

$$(b) \quad G(s) = \frac{V_o(s)}{V_s(s)}$$



$$\therefore V_a = \frac{1A_1}{1A}, \quad V_b = \frac{1A_2}{1A}$$

$$V_o = V_a - V_b, \quad G = \frac{V_o}{V_s}$$

$$G(s) = \frac{(R_1 R_2 R_3 C - L R_3) s}{(2 R_1 R_2 C + L R_1 R_3 C) s^2 + (L R_1 + 2 R_2 + 2 R_3 + R_1 R_2 R_3 C) s + R_1 R_2 + R_2 R_3}$$

2. Rotational Mechanical Systems (25 marks) (By hand & By MATLAB)

A DC motor is used to control the steering angle in a self-driving car. A simplified schematic of the setup is shown in the figure below. The motor has resistance R with negligible inductance, torque constant K_t , and back emf constant K_e . The motor's rotor, with moment of inertia J_m , is connected via a shaft with inertia J_s and rotational spring k_s to a set of gears and to the steering rack and pinion with an equivalent inertia of J_p . The resistance resulting from turning the wheels is represented using a rotational spring of stiffness k_p and a damper b_p .

Note: Θ_m is positive in the clockwise direction. You should be able to figure out all the other directions from this information.

Note: A warning in advance - the algebraic substitutions can get really messy. You should use MATLAB to help with simplifying the expression and solving for determinants. You must show all matrices and vectors by hand to receive full marks for this question.

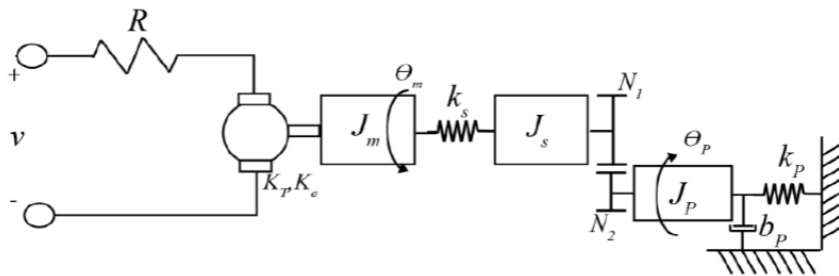


Figure 2: Simplified Automated Steering System.

- (a) (2 marks) Derive the 'relationship' for the electrical system relating the applied voltage $v(t)$, current and the motor rotation Θ_m (do not consider the rotational system here).

- (b) (15 marks) Derive the transfer function for the output of the pinion rotation Θ_p from the input motor torque τ_m .
- (c) (8 marks) Derive the overall system transfer function (i.e., with $V(s)$ as input and $\Theta_p(s)$ as output).

Q2.

$$(a) V(s) - I(s) \cdot R - k_b s \Theta_m = 0, \quad T_m = k_t I(s)$$

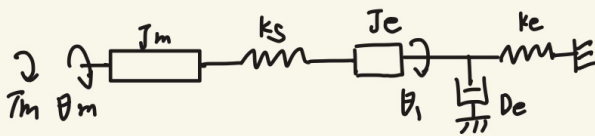
$$\rightarrow V - \frac{T_m}{k_t} R - k_b s \Theta_m = 0$$

$$\rightarrow T_m(s) = - \frac{k_b k_t}{R_a} \omega_m + \frac{k_t}{R_a} e_a$$

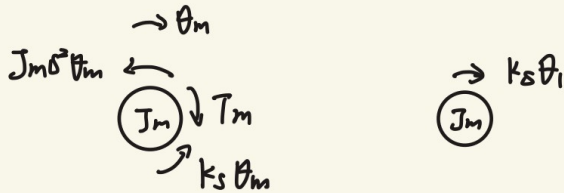
$$(b) G(s) = \frac{\Theta_p}{T_m}$$

$$(1) \quad \frac{T_m}{k_t} R + k_b s \Theta_m = V$$

2. Equivalent to N_1 axis

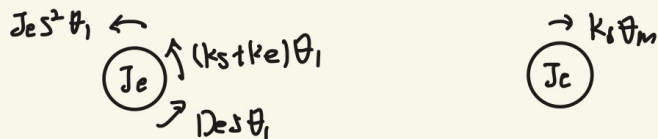


$$J_c = J_s + \frac{N_1^2}{N_2^2} J_p, \quad D_e = \frac{N_1^2}{N_2^2} b_p, \quad k_e = \frac{N_1^2}{N_2^2} k_p$$



$$\therefore J_m s^2 \Theta_m + k_s \Theta_m = T_m + k_s \Theta_1$$

$$(J_m s^2 + k_s) \Theta_m - k_s \Theta_1 = T_m \quad (1)$$



$$\therefore J_e s^2 \theta_1 + D_e s \theta_1 + (k_s + k_e) \theta_1 = k_s \theta_m$$

$$\therefore -k_s \theta_m + (J_e s^2 + D_e s + k_s + k_e) \theta_1 = 0 \quad (3)$$

$$\begin{bmatrix} J_m s^2 + k_s & -k_s \\ -k_s & J_e s^2 + D_e s + k_s + k_e \end{bmatrix} \begin{bmatrix} \theta_m \\ \theta_1 \end{bmatrix} = \begin{bmatrix} T_m \\ 0 \end{bmatrix}$$

$$\therefore \theta_1 = \frac{|A_2|}{|A|} \quad \therefore \frac{\theta_p}{\theta_1} = \frac{N_1}{N_2} \quad \therefore \theta_p = \frac{N_1}{N_2} \theta_1$$

$$G(s) = \frac{\theta_p}{T_m} = \frac{N_1 N_2 k_s}{(J_m J_p N_1^2 + J_m J_s N_2^2) s^4 + (J_m N_1^2 b_p) s^3 + (J_m N_1^2 k_p + J_m N_2^2 k_s + J_p N_1^2 k_s + J_s N_2^2 k_s) s^2 + (N_1^2 b_p k_s) s + N_1^2 k_p k_s}$$

$$(c) G(s) = \frac{\theta_p}{V}$$

$$\text{Note: } T_m = \frac{-k_e k_T}{R_a} s \theta_m + \frac{k_T}{R} V, \quad V = \frac{T_m}{k_T} R + k_e s \theta_m$$

$$\theta_m = \frac{\det(A_1)}{\det(A)}$$

$$G(s) = \frac{\theta_p}{V} =$$

3. Electrical Circuits and Mechanical Systems (40 marks) (By hand & Check By MATLAB)

A DC motor is connected to an op-amp circuit in cascade as shown in Figure 3a. The op-amp circuit is shown in Figure 3b; the input to the op-amp is a voltage source (battery) $v_i(t)$, the output is the voltage $v_o(t)$, and the transfer function of this subsystem is $G_1(s)$. The DC

motor subsystem is shown in Figure 3c; the input to the DC motor is the op-amp's output $v_s(t)$, the output is the angular velocity $\omega(t)$ of a shaft connected to the motor and the transfer function of this subsystem $G_2(s)$. The DC motor subsystem is not loading the op-amp circuit subsystem. We will analyze the behaviour of each subsystem independently, and that of the overall cascaded system.

Note: $\omega(t)$ is positive in the clockwise direction. You should be able to figure out any other directions from this information.

Note: The algebraic manipulations here are easy, you can use Matlab to solve for your transfer functions and sub in the provided values but you do not have to.

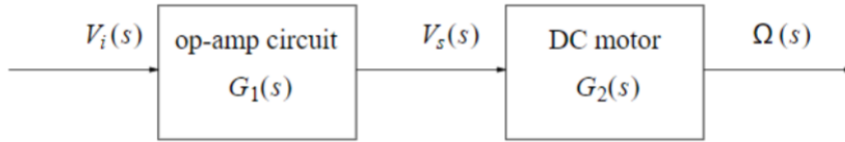


Figure 3a: Cascade of an op-amp circuit subsystem and a DC motor subsystem.

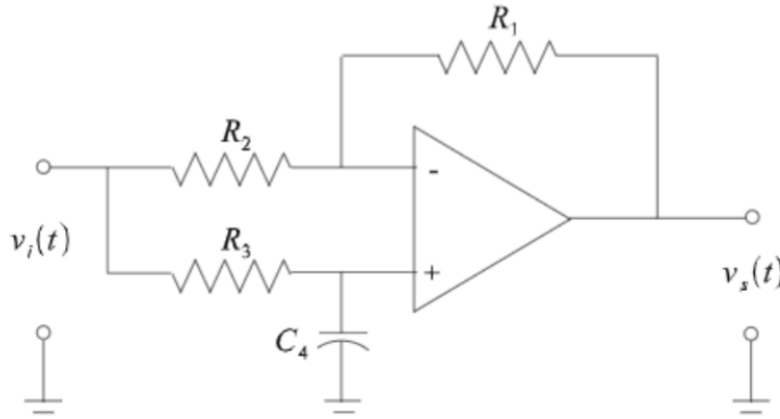


Figure 3b: The op-amp circuit subsystem.

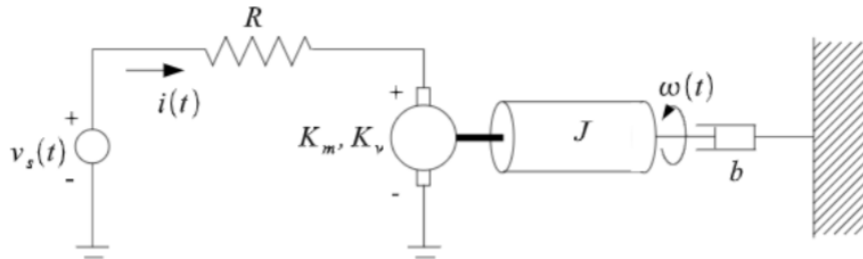


Figure 3c: The DC motor subsystem.

Additional data for the op-amp circuit subsystem are: the resistors $R_1 = 100\Omega$, $R_2 = 100\Omega$, and $R_3 = 100k\Omega$, and the capacitor $C_4 = 1\mu F$.

Additional data for the DC motor subsystem are: motor resistance $R = 1\Omega$; the torque constant $K_m = 1N \cdot m/A$ and back-emf constant $K_v = 1V \cdot sec/rad$; the load shaft inertia $J = 1kg \cdot m^2$ and viscous friction coefficient $b = 1N \cdot m \cdot sec/rad$. The motor's own inertia and inductance are negligible.

- (10 marks) Derive the transfer function $G_1(s)$ of the op-amp circuit subsystem.
- (10 marks) Derive the transfer function $G_2(s)$ of the DC motor circuit subsystem.
- (5 marks) Derive the overall system transfer function $G(s)$ with $V(s)$ as input and $\Omega(s)$

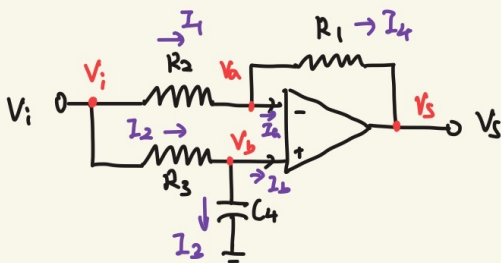
- (c) (5 marks) Derive the overall system transfer function (i.e., with $v_i(s)$ as input and $v_o(s)$ as output), we will call this $G_3(s)$.
- (d) (5 marks) Determine the steady-state $\omega(t)$ when the input $v_i(t)$ is a step function of amplitude 10Volts (i.e., the unit-step response). Can you apply Final Value Theorem here? Why or why not?
- (e) (5 marks) Derive the time-domain response $\omega(t)$ when the input $v_i(t)$ is a step function of amplitude 1Volt (i.e., the unit-step response). Use Table 2.1 and Table 2.2 of the textbook (given at the end of this paper).
- (f) (5 marks) Sketch the time-domain response $\omega(t)$ you obtained in part (e). Does this match your answer in part (d)? Why or why not?

Q3.

$$R_1 = 100 \Omega, R_2 = 100 \Omega, R_3 = 100 \times 10^3 \Omega, C_4 = 1 \times 10^{-6} F, R = 1 \Omega$$

$$K_m = 1 V/m/A, K_v = 1, J = 1, b = 1$$

(a) $G_1(s) = \frac{V_o}{V_i}$



$$V_a = V_b, I_a = I_b = 0. \quad \therefore I_1 = I_a + I_4 = I_4 = V$$

$$I_2 = I_3 + I_b = I_3$$

$$\therefore \frac{V_i - V}{R_2} = \frac{V - V_o}{R_1} \rightarrow \frac{V_i - V}{100} - \frac{V - V_o}{100} = 0$$

$$\frac{V_i}{100} + \frac{V_o}{100} = \frac{V}{50} \quad (1)$$

$$\frac{V_i - V_b}{R_3} = \frac{V_b}{\frac{1}{Cs}} \rightarrow \frac{V_i}{R_3} - \frac{V}{R_3} - Cs V = 0$$

$$\frac{V_i}{1 \times 10^5} = \left(\frac{1}{10^5} + 1 \times 10^{-6} s \right) V \quad (2)$$

$$\begin{bmatrix} \frac{1}{10s} & \frac{1}{10s} \\ \frac{1}{10s} & 0 \end{bmatrix} \begin{bmatrix} V_i \\ V_s \end{bmatrix} = \begin{bmatrix} \frac{V}{50} \\ (\frac{1}{10s} + 10^{-b_s})V \end{bmatrix}$$

$$V_i = \frac{\det(A_1)}{\det(A)} \quad , \quad V_s = \frac{\det(A_2)}{\det(A)}$$

$$G_1 = \frac{V_s}{V_i} = \frac{10-s}{s+10}$$

$$(b) G_2(s) = \frac{\Omega(s)}{V_s(s)}$$

1. Circuit: $K_b = K_v = 1$, $K_m = K_e = 1$, $T_m(s) = K_e I_a(s)$, $V_b(s) = K_b s \theta_m$

$$V_s(s) - I_a(s) \cdot R - K_v s \theta_m = 0 \rightarrow V_s(s) - \frac{T_m(s)}{1} - s \theta_m = 0$$

$$\therefore T_m(s) + s \theta_m = V_s(s) \quad \textcircled{1} \rightarrow T_m = V_s - s \theta_m = V_s - \Omega_m$$

2. mechanical:

$$\begin{array}{c} \rightarrow \theta_m \\ \nwarrow J s^2 \theta_m \\ T_m \nearrow \textcircled{J} \searrow b s \theta_m \end{array} \quad \therefore J s^2 \theta_m + b s \theta_m = T_m$$

$$\therefore s^2 \theta_m + s \theta_m = T_m \rightarrow s \Omega_m + \Omega_m = T_m = V_s - \Omega_m$$

$$\therefore s \Omega_m + 2 \Omega_m = V_s$$

$$G_2(s) = \frac{\Omega(s)}{V_s(s)} = \frac{1}{s+2}$$

$$(c) G_3(s) = \frac{\Omega(s)}{V_i(s)} = G_1 G_2 = \frac{V_s}{V_i} \cdot \frac{\Omega}{V_s} = \frac{10-s}{s+10} \cdot \frac{1}{s+2} = \frac{10-s}{(s+10)(s+2)}$$

$$(d) \lim_{t \rightarrow \infty} w(t) \quad , \quad V_i(t) = u(t) \cdot 10 \quad , \quad V_i(s) = \frac{10}{s}$$

$$\lim_{t \rightarrow \infty} w(t) = \lim_{s \rightarrow 0} s \Omega(s) \quad , \quad \Omega(s) = G_3(s) V_i(s) = \frac{100-10s}{s(s+10)(s+2)}$$

$\therefore$ Can use FVT

$$\therefore \lim_{t \rightarrow \infty} w(t) = \frac{100}{20} = 5$$

(e) $w(t) = ?$ $v_i(t) = u(t)$, $V_i(s) = \frac{1}{s}$

$$\Omega(s) = G_3 V_i = \frac{10-s}{s(s+10)(s+2)}$$

$$w(t) = \mathcal{L}^{-1}[\Omega(s)] , \quad \Omega(s) = \frac{A}{s} + \frac{B}{s+10} + \frac{C}{s+2}$$

$$A = \frac{10-s}{(s+10)(s+2)} \Big|_{s=0} = \frac{10}{20} = \frac{1}{2}$$

$$B = \frac{10-s}{s(s+2)} \Big|_{s=-10} = \frac{20}{-10 \times (-8)} = \frac{1}{4}$$

$$C = \frac{10-s}{s(s+10)} \Big|_{s=-2} = \frac{12}{-2 \times 8} = -\frac{12}{16} = -\frac{3}{4}$$

$$\therefore w(t) = \mathcal{L}^{-1}\left[\frac{1}{2} \frac{1}{s} + \frac{1}{4} \frac{1}{s+10} - \frac{3}{4} \frac{1}{s+2}\right]$$

$$= \frac{1}{2} u(t) + \frac{1}{4} e^{-10t} u(t) - \frac{3}{4} e^{-2t} u(t)$$

(f) Does not match since inputs are different.

4. Linearization (10 marks) (By hand)

Many systems are piecewise linear. That is, over a large range of variable values, the system can be described linearly. A system with amplifier saturation is one such example. Given the

can be described linearly. A system with amplifier saturation is one such example. Given the differential equation:

$$\frac{d^2x}{dt^2} + 17\frac{dx}{dt} + 50x = f(x)$$

assume that $f(x)$ is as shown in Figure 4 below. Write the differential equation and associated Laplace equation for each of the following ranges of x :

- (a) $-\infty < x < -3$
- (b) $-3 < x < 3$
- (c) $3 < x < \infty$

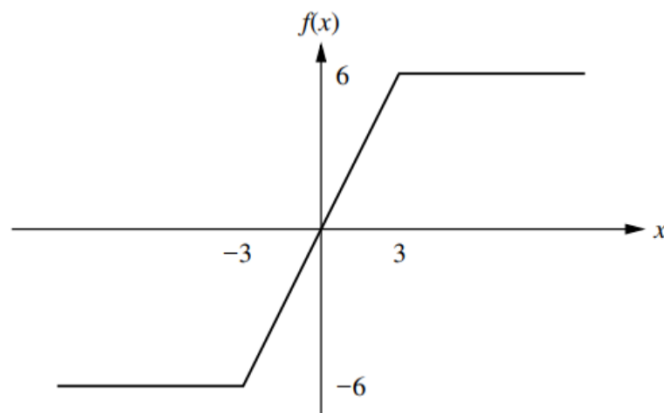


Figure 4: Amplifier range.

Q4.

$$(a) \quad \frac{d^2x}{dt^2} + 17\frac{dx}{dt} + 50x = -6$$

$$s^2X(s) + 17sX(s) + 50X(s) = -\frac{6}{s}$$

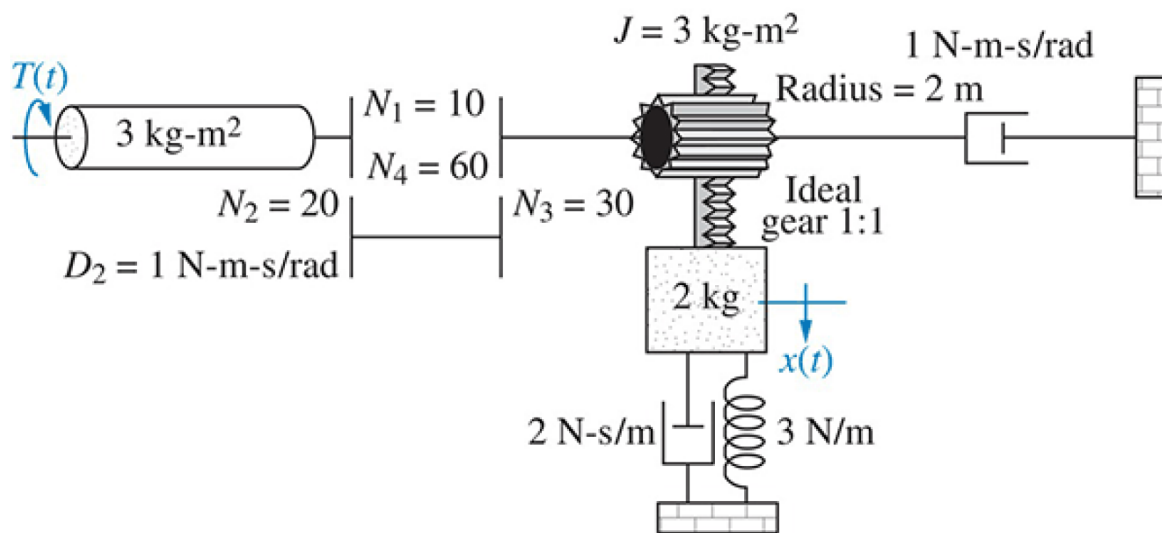
$$(b) \quad \frac{d^2x}{dt^2} + 17\frac{dx}{dt} + 50x = 2x$$

$$s^2X(s) + 17sX(s) + 50X(s) = 2X(s)$$

$$(c) \quad \frac{d^2x}{dt^2} + 17\frac{dx}{dt} + 50x = 6$$

$$s^2X(s) + 17sX(s) + 50X(s) = \frac{6}{s}$$

33. For the combined translational and rotational system shown in [Figure P2.21](#), find the transfer function, $G(s) = X(s)/T(s)$.
[\[Sections: 2.5; 2.6; 2.7\]](#)



Q33 1 DOF

$$T_c: \because \frac{T_1}{T_2} = \frac{N_1}{N_2} \therefore T_2 = \frac{N_2}{N_1} T_1 \because \frac{T_2}{T_3} = \frac{N_3}{N_4} \therefore T_3 = \frac{N_4}{N_3} T_2$$

$$\therefore T_e = T_3 = \frac{N_4}{N_3} \cdot \frac{N_2}{N_1} T = \frac{60}{30} \times \frac{20}{10} = 2 \times 2 \times 7 = 4 \text{ T(s)}$$

$J_1 (3 \text{ kg m}^2)$ equivalent to N_4 axis:

$$J_{1e} = J_1 \cdot \frac{N_2^2}{N_1^2} \cdot \frac{N_4^2}{N_3^2} = 3 \times 4 \times 4 = 48$$

$m_1 (2 \text{ kg})$ equivalent to N_4 rotation axis:

$$\therefore F = am = s^2 \chi m, \quad T = rF = Js^2 \theta = r(s^2 \chi m) = rs^2 r \theta m \\ = (r^2 m) s^2 \theta$$

$$\therefore J_{m1} = r^2 m = 4 \times 2 = 8$$

$$f_d = 25 \chi(s), \quad f_s = 3 \chi(s) \quad \therefore T_d = r f_d = 45 \chi(s) = 45 r \theta = 85 \theta$$

$$T_s = r f_s = 6 \chi(s) = 6 r \theta(s) = 12 \theta(s)$$

$$\therefore J_e = J_{m1} + J_{1e} + 3 = 8 + 48 + 3 = 59$$

$$D_{2e} = D_2 \frac{N_4^2}{N_3^2} = 4$$

$$\therefore 59 s^2 \theta + 15 \theta + 85 \theta + 12 \theta + 45 \theta = 4T$$

$$\therefore 59 s^2 \theta + 135 \theta + 12 \theta = 4T$$

$$\therefore \frac{\theta(s)}{T(s)} = \frac{4}{59 s^2 + 135 s + 12}$$

$$\therefore \chi(s) = r \theta(s) \quad \therefore \frac{\chi(s)}{T(s)} = r \frac{\theta(s)}{T(s)} = \frac{8}{59 s^2 + 135 s + 12}$$